

Sprint 1 — Secure Simulation Considerations

Capstone Team Project (B)

1 Introduction

Through the progression of the E-Waste Battery Extraction project the research and development processes may expose user devices such as seen with ROS2 and MoveIt exposure to the entire network without appropriate authentication methods, or sharing of files. Additionally, members may become lack with ensure the legitimacy of files and forget to ensure best practices are present throughout their roles.

This document aims to ensure best practices and methods are placed forward for current and future team members to follow so that development of the e-waste battery extraction be completed so safely.

2 Pretense

The requirements for successful simulation to occur must be understood prior to the creation of a useable system or methodology.

2.1 Requirements

The following are minimal requirements for consideration if a solution were to be implemented for a secure environment for simulation.

1. The solution must allow for either simple download or running of the simulation safely and securely.
2. The solution must not overreach the 100 dollar budget allocated to the project. If possible preferable the solution must cost nothing or is for free.
3. The solution must be easily manageable and accessible to all members while still ensure security is placed.
4. The solution must allow Linux as ROS2/MoveIt2 only allow for Linux to operate.
5. The solution must either be heavily documented or simple to manage for future students — This may be flexible as the solution though complex can also be intuitive for easier pick-up.

The following are system requirements which **do not** highlight best practices that allow team members must follow. As it should be understood the separation between best practices and system requirements differ in their actions

3 Purposed System

The primary unique simulation environments are to be considered. The first instance in which the simulation machine hosted via a dedicated singular remote able desktop either physically located through another student or within Deakin University.

The second option is the consideration of a cloud desktop that would fall under the budget constraints of the project, which will required dedicated network administrators to manage throughout each trimesters

In either option ownership of the initial instances in integral to ensure continuous security and access for users. The locally stored remote able desktop may be a configuration installation on hardware which could occur on a singular students device, selected each trimester of the project. This would allow for the creation of a shared environment with dedicated hardware to ensure smooth operations, accessible via the secure remote application Tailscale.

In either option the purpose system of either a shared singular device securely accessible either via individual different login accounts or shared keys. User should **never** share the same login information.

3.0.1 TailScale

Tailscale is an application that allows remote lobe access to an application utilising tailscale on both ends. This options allows for a remote connection without the requirement to adjust router configurations and isolates to the device, now exposing a users entire network.

4 Interim

In the as a dedicated system cannot be brought or utilise members working with ROS2 and MoveIt2 should be aware that the usage of nodes for the control programs contain no authentication process to ensure legitimate connection between either node. This presents the security concerns of a malicious user either within the school campuses, over a shared network, or an area in which a developers network is shared to send malicious code towards the user node accessing sending arbitrary code.

4.1 Mitigation

To mitigate the potential threat students should encapsulate their virtual machines within an isolated network. Utilising the virtual machine solely for programming and running the Python code for simulation. All relevant searches or importation of files should be completed via the host machine web-browser and a connected shared folder between the host and virtual machine.

5 Conclusion

While development of the simulation environment is integral for the project students should still ensure they are completing their tasks safely as to not cause harm towards themselves or the project as a result of a malicious threat actor.